

Intro to Hilbert Spaces

Idea: Build up functions from linear combos.

$$(*) \quad f(x) \approx c_1 e_1(x) + c_2 e_2(x) + \dots + c_n e_n(x)$$

Question 1: How to select $c_1, \dots, c_n$?

Question 2: How to analyze accuracy " $\approx$ "?

Goal: Develop "linear algebra for functions" in order to answer questions 1 & 2.

$\Rightarrow$ A vector space is a nonempty set V w/ a scalar field ($\mathbb{R}$ or $\mathbb{C}$), closed under addition and scalar multiplication.

$\Rightarrow$ A set $\{e_1, \dots, e_n\}$ is a basis for V if it spans V and is linearly independent.

$$x \in V \quad \Leftrightarrow \quad x = \overset{\substack{\text{unique to } x \\ \downarrow}}{c_1} e_1 + \dots + \overset{\substack{\text{unique to } x \\ \downarrow}}{c_n} e_n$$

Norms : Inner Products

Given a vector space V , a **norm** on V is a map $\|\cdot\|: V \rightarrow [0, \infty)$ that satisfies

$$i) \|x\| \geq 0, \text{ and } \|x\| = 0 \text{ IFF } x = 0.$$

$$ii) \|\alpha x\| = |\alpha| \|x\|$$

$$iii) \|x+y\| \leq \|x\| + \|y\|$$

(or $\mathbb{R}$)

An **inner product** on V is a map $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{C}$ that satisfies the following criteria:

$$i) \langle x, y \rangle = \overline{\langle y, x \rangle}$$

$$ii) \langle \alpha x, y \rangle = \alpha \langle x, y \rangle$$

$$iii) \langle x_1 + x_2, y \rangle = \langle x_1, y \rangle + \langle x_2, y \rangle$$

$$iv) \langle x, x \rangle \geq 0 \text{ and } [= 0 \text{ IFF } x = 0]$$

Question: What is $\langle x, \alpha y \rangle$ in terms of $\langle x, y \rangle$?

$$\Rightarrow \langle x, \alpha y \rangle = \overline{\langle \alpha y, x \rangle} = \overline{\alpha \langle y, x \rangle} = \overline{\alpha} \overline{\langle y, x \rangle} = \overline{\alpha} \langle x, y \rangle$$

$\Rightarrow$ "conjugate linear" in second argument

Example: $V = \mathbb{C}^n$ $\langle x, y \rangle = y^* x = \bar{y}_1 x_1 + \bar{y}_2 x_2 + \dots + \bar{y}_n x_n$

Example: $V = C[-1, 1]$ $\langle f, g \rangle = \int_{-1}^1 f(x) \overline{g(x)} dx$

Inner Product Spaces

A vector space V with inner product $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{C}$ (or $\mathbb{R}$) is an inner product space.

The inner product induces a norm $\|\cdot\|: V \rightarrow \mathbb{R}_+$

$$\|x\| = \sqrt{\langle x, x \rangle} \quad \text{for each } x \in V.$$

Two vectors $x, y \in V$ are orthogonal if $\langle x, y \rangle = 0$ and we write $x \perp y$.

Note that if $x \perp y$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

$$\begin{aligned} \text{p.f.} \quad \langle x+y, x+y \rangle &= \langle x, x \rangle + \langle y, x \rangle + \langle x, y \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 0 + 0 + \|y\|^2 \end{aligned}$$

The Cauchy-Schwarz inequality allows us to control the size of inner products w/norms:

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Example: Consider the inner product

$$\langle f, g \rangle = \int_{-1}^{+1} f(x) \overline{g(x)} dx.$$

$\Rightarrow$ The induced norm is $\|f\| = \left(\int_{-1}^{+1} |f(x)|^2 dx \right)^{1/2}$.

$\Rightarrow$ The Cauchy-Schwarz inequality reads

$$\left| \int_{-1}^{+1} f(x) \overline{g(x)} dx \right| \leq \left(\int_{-1}^{+1} |f(x)|^2 dx \right)^{1/2} \left(\int_{-1}^{+1} |g(x)|^2 dx \right)^{1/2}$$

Question: What are some examples of orthogonal pairs of functions?

$\Rightarrow$ If f is odd and g is even, then $\int_{-1}^{+1} x^3 dx = \frac{x^4}{4} \Big|_{-1}^{+1} = 0$

$$\begin{aligned} \int_{-1}^{+1} f(x) \overline{g(x)} dx &= \int_{-1}^{+1} h(x) dx = \int_{-1}^0 h(x) dx + \int_0^1 h(x) dx \\ &= \int_0^1 h(-x) (-dx) + \int_0^1 h(x) dx = \int_0^1 h(-x) + h(x) dx = 0 \end{aligned}$$

odd $h(x) = -h(-x)$
 $x \rightarrow -x, dx \rightarrow -dx$

$\Rightarrow$ If $f(x) = \cos n\theta$ and $g(x) = \cos 2n\theta$

$$\int_{-1}^{+1} \cos n\theta \cos 2n\theta d\theta = \int_{-1}^{+1} \frac{1}{2} (e^{in\theta} + e^{-in\theta}) \frac{1}{2} (e^{i2n\theta} + e^{-i2n\theta}) d\theta$$

$$\begin{aligned}
&= \frac{1}{4} \int_{-1}^1 e^{i3n\theta} d\theta + \frac{1}{4} \int_{-1}^1 e^{-in\theta} d\theta + \frac{1}{4} \int_{-1}^1 e^{in\theta} d\theta + \frac{1}{4} \int_{-1}^1 e^{-i3n\theta} d\theta \\
&= \frac{1}{4} \left[\frac{e^{i3n\theta}}{3ni} \Big|_{-1}^1 + \frac{e^{-in\theta}}{-ni} \Big|_{-1}^1 + \frac{e^{in\theta}}{ni} \Big|_{-1}^1 + \frac{e^{-i3n\theta}}{-3ni} \Big|_{-1}^1 \right] \\
&= 0
\end{aligned}$$

Hilbert Spaces

A (real or complex) Hilbert space H is

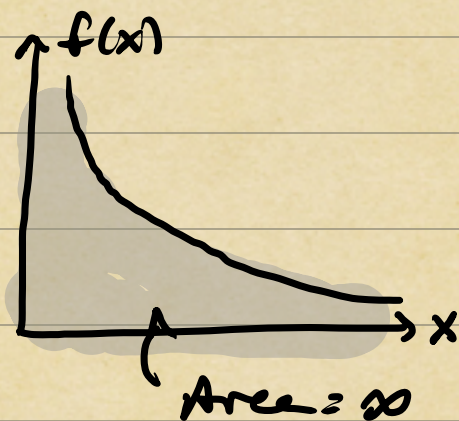
- i) a vector space (over $\mathbb{R}$ or $\mathbb{C}$)
- ii) equipped with an inner product $\langle \cdot, \cdot \rangle$
- iii) complete (Cauchy sequences converge)

Completeness

In normed inner product spaces of functions, we must wrestle with the fact that some functions have "infinite" length.

Example: $f(x) = 1/\sqrt{x}$

$$\begin{aligned}\|f\|^2 &= \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon}^1 \frac{dx}{x} = \lim_{\varepsilon \rightarrow 0} \left. \frac{-1}{x} \right|_{\varepsilon}^1 \\ &= \infty\end{aligned}$$



Therefore, we typically restrict to $\{f: \|f\| < \infty\}$.

We now face the question of which ^{infinite} linear combinations we will allow in our function space. The basic idea of completeness is that we should allow all ^{infinite} linear combinations which converge to something with finite norm.

Consider a vector space V w/ inner product $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{C}$ and induced norm $\| \cdot \| = \sqrt{\langle \cdot, \cdot \rangle}$.

A sequence $v_1, v_2, v_3, \dots \in V$ is *Cauchy* if for every $\varepsilon > 0$, there is an N s.t. for all $m, n > N$

$$\|v_m - v_n\| < \varepsilon.$$

A complete space^V is one in which every Cauchy sequence converges to an element of V .

For example, suppose that the linear combos

$$v_n = \sum_{k=1}^n a_k f_k \quad \begin{matrix} a_k \in \mathbb{C} \\ f_k \in V \end{matrix}$$

form a Cauchy sequence. Then, completeness requires that there is an element $f \in V$ s.t.

$$\lim_{n \rightarrow \infty} \|v_n - f\| = 0.$$

Therefore, we can identify f with the infinite linear combination (series)

$$f = a_1 f_1 + a_2 f_2 + \dots + a_n f_n + \dots = \sum_{k=1}^{\infty} a_k f_k.$$

In particular, any series that converges absolutely in the Hilbert space norm, i.e.,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \|a_k f_k\| = 0$$

is Cauchy, and, therefore, converges to some limit in V : there is an $f \in V$ s.t.

$$f = \lim_{n \rightarrow \infty} \sum_{k=1}^n \alpha_k f_k.$$